

Tobi Oloke

Software Engineer // Haskell Enthusiast

Southampton, UK

tobi@tobioloke.com | <https://tobioloke.com> | <https://github.com/tobz619>

I am an enthusiastic, passionate, self-taught programmer and software engineer with an desire for learning the fundamental ground truths and implementing the simplest and most effective solutions where possible. I like being challenged and absorbing and applying myself to new knowledge. Even though I am only approaching 5 years of experience in all things SWE, I've come a long way and know I'm in it for the long haul and I'm extremely keen to help you solve your problems and build your solutions!

Key Skills

- Programming Languages: Python (Proficient), C/C++ (Proficient), Haskell (Native), Nix (Proficient), Typescript (Intermediate), SQL (Proficient).
- Development Methodologies: Agile, Waterfall, Kanban
- Version Control: Git, Nix flakes, Docker
- DevOps / CI/CD: Jenkins, Nix
- Operating Systems: Windows, Unix (Ubuntu, NixOS), macOS
- Soft Skills: Teamwork, Communication, Problem-Solving, Time Management

Extra-Curricular

- Building a real-time 3D renderer in Haskell as a learning project.
- Co-maintainer of the site <https://boardword.com>.
- Member of the Haskell Foundation Exam Board testing and feedback group.
- Volunteer teacher at a local school for "Computing" skills including programming, data structures, algorithm design, robotics etc.

Recent Relevant Work Experience

Software Engineer @ Roke Manor Research (September 2022 - Present)

Notable contributions during my tenure here include:

- Maintainer of a formerly deprecated DPDK/C++ networking stack which filters and processes packets in real time after producing a plan to upgrade it to newer OS and kernel versions. (03/26-Present)
- Maintainer of a C++ and Python platform which, after reverse engineering firmware, creates tooling to manipulate and provide desirable properties. Implementation of CI/CD pipeline using Jenkins + Gitlab. Refactoring of C++ codebase using functional programming fundamentals such as scoping functions and even stronger typing (09/2025 - Present)
- Creating Python, bash and C-based tools from Haskell-like design patterns for the static analysis of applications which dropped analysis times by more than 50%, improved documentation and provided an additional product line to the customer. (01/24-04/24, 06/24-09/25). The tools include:
 - Android device automation, forming a product that automates the setup of Android devices and the applications on them to a target, idempotent state.

- A tool to automatically generate standardised documentation that can be viewed on screens of arbitrary sizes and documents of arbitrary length using plain html.
- Contributed to a natural language processing platform for the direction of automated agents by writing underlying actions specs and algorithms to carry out instructions. (04/24-Present)
- The hierarchical planning model leverages constraint programming to dynamically and automatically generate satisfactory plans from text input which does not violate constraints and uses existing instructions to create emergent behaviour.
- Developed a prototype real-time “Predictor” for mobile telecommunications messages types based on Haskell Arrow design in Python. The goal was to figure out which message would come next in a sequence of messages like “Predictive text”. In reality, it was effectively a directed graph which kept a short-term context window of the seen messages and used that context to then predict the next node via a regression function: the more data, the more accurate it became. (12/23 - 01/24)
- Assisted in the creation of a semi-automated system for the provisioning and command and control of unix devices using Ansible(10/23-11/23).
- Worked with C, Typescript and Python to develop a suite of tools for monitoring embedded networking firmware with an eye towards modifying and augmenting capabilities.
- Prototyped an “immune system” for linux devices by creating a database of syscall sequences for normal behaviour and testing against non-standard behaviour. This also leveraged my biological background. (09/23 - 10/23).
- Designed and implemented modules for a suite of capability for interlinked programs and capabilities mainly in Python. (03/23 - 09/23)
- Enabled custom embedded linux on devices and designed capability for them. (01/23 - 04/23)
- Investigated smart contract cryptocurrencies and created strategies for developing smart contracts. (10/22 - 01/23)
- Developed a Python Wrapper to enable Android device automation. (11/22 - 01/23)

Project Coordinator @ Lifegate Communities Walsall for BBC Children In Need (Summers 2017 - 2022)

- Successfully devised plans and activities for children as well as led a team of 10 volunteers managing up to 50 children for 10 days for the BBC Children In Need communities project. This was a successful project which has been renewed and expanded year on year.
- Managed staffing and payroll via carefully crafted Excel spreadsheets and handled sensitive staff and child data with preliminary user-access control from 2019 onwards.

Education

University of Helsinki, Haskell Massive Online Open Course (Jan 2022 - May 2022)

I achieved full marks in this course, self-teaching myself Haskell and working through the exercises found within this course and thoroughly enjoyed it! My answers can be accessed here:

<https://github.com/tobz619/Haskell-MooC-Solutions>

University of Derby, Applied Sport and Exercise Science (Physiology) MSc (2019 -2020)

University of Leicester, Medical Physiology BSc (2016 -2019)

Queen Mary’s Grammar School (2009 - 2016)